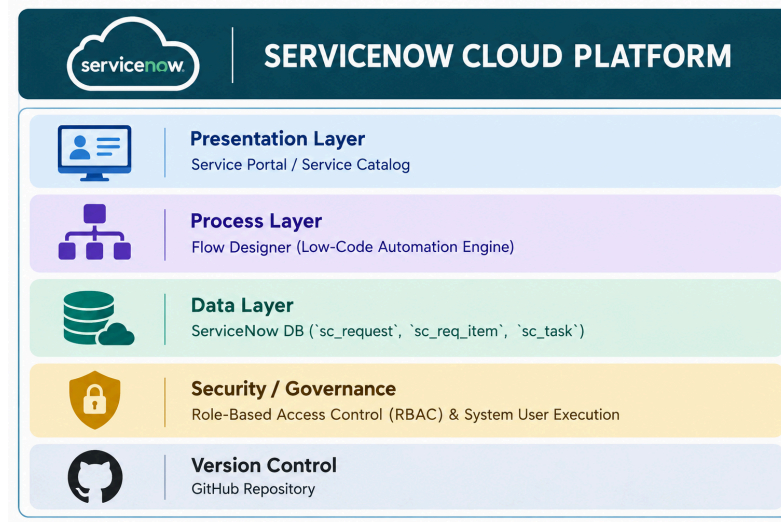


Project Design Phase-II Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	SWTID-2026-6819
Team Leader	Hazna Mariyam. H
Team Members	Anuskha. K Theras Jasmine. M
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

1. Overview & Technology Stack Summary

The "Streamlining IT Procurement" solution leverages ServiceNow's cloud-native, low-code architecture to automate hardware task creation and assignment upon service request approval.



2. Core Platform Components & Modules

Module / Component	ServiceNow Tech / Table	Function & Purpose
Automation Engine	Flow Designer (sys_hub_flow)	Replaces legacy script workflows with a visual, maintainable flow model (Standard laptop task).
Service Catalog	Maintain Items (sc_cat_item)	Hosts the Standard Laptop item, providing user input forms and checkout capabilities.
Data Store (Requests)	Requested Item Table (sc_req_item)	Stores individual catalog order details and status triggers for execution.
Data Store (Tasks)	Catalog Task Table (sc_task)	Stores dynamically created fulfillment tasks for the Hardware engineering group.

User & Group Admin	User Group Table (sys_user_group)	Manages the target assignment group (Hardware) for automated task routing.
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3. Technology & Architecture Specifications

3.1 Application Scope & Security

- **Application Scope:** Global
- **Execution Authority:** System User (Ensures executions bypass user-level ACL limitations during automated background task creation).
- **Role-Based Access Control (RBAC):**
 - admin / catalog_admin: Flow creation, process engine binding, catalog configuration.
 - itil: Accessing and completing generated Catalog Tasks (sc_task).
 - snc_internal: End-user capability to order hardware via the Service Portal.

3.2 Key System Interfaces & Data Flow

- **Trigger Interface:** Service Catalog Event -> Flow Engine.
- **Data Mapping Mechanism:** Data Pill Drag-and-Drop references (Trigger -> Requested Item Record).
- **Field Values Engine:** Native Flow Designer Action (Create Catalog Task).

4. Environment & Tooling Prerequisites

Category	Tool / Environment	Requirement & Purpose
Cloud Platform	ServiceNow Developer Instance (PDI)	Deployment environment for Flow Designer and Service Catalog configuration.
Web Browser	Google Chrome / Microsoft Edge	Administrative interface for ServiceNow Studio & Flow Designer.
Version Control	GitHub	Repository hosting project documentation across all 7 phases.
Development Environment	ServiceNow Flow Designer UI	Low-code designer for building, testing, and activating business workflows.